The thermohydraulic behavior of smooth twisted tape inserts utilizing oxide nanofluids in turbulent flow within tubes.

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A Project Report

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CERTIFICATE

It is certified that the work contained in this project report entitled "**Thermohydraulic Performance of Plain Twisted Tape Inserts with Oxide Nanofluids in Turbulent Tube Flow**" is submitted for the course ME1406: Minor Project, Spring semester 2026. This work has been carried out under the supervision of **Dr. Laljee Prasad** and this work has not been submitted elsewhere for a degree.

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This serves to formally certify that the technical declarations and empirical assertions articulated by the candidates herein are substantiated and fundamentally accurate to the best of my professional knowledge.

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Abstract

Among the three nanofluids investigated, GO/water shows the highest Nusselt number enhancement, achieving up to approximately 34% improvement over water-based plain twisted tape at TR = 2, followed by CuO/water at approximately 27% and Al_2O_3/water at approximately 22%. This ordering reflects the inherently superior thermal conductivity of graphene oxide relative to the other two oxide nanofluids, and remained consistent across all tested Reynolds numbers and volume concentrations under turbulent flow conditions (Re = 10,000–22,000). Higher volume concentrations (0.5%) yielded greater thermal enhancement, though accompanied by a corresponding pressure drop penalty. Lower twist ratios, particularly TR = 2, produced the highest Nusselt numbers and friction factors owing to the increased swirl intensity generated by the more tightly wound tape geometry, while thermohydraulic performance (η) peaked at TR = 3 for GO/water, indicating that the pressure drop penalty at TR = 2 partially offsets the thermal benefit when overall system efficiency is considered.

When we compare the results to a regular tube and the hyperbolic-cut twisted tape data from Rajan et al, we find something really interesting. It turns out that using a simple twisted tube with a special kind of fluid, called GO/water nanofluid, gives us performance that's similar to the more complex hyperbolic-cut twisted tape when it's used with regular water. This means that we might be able to get the same benefits from using a simpler design with the right kind of fluid, rather than having to use a more complicated design. This could make the simpler design a more cost-effective option for heat exchanger tubes, especially when we use high-conductivity nanofluids like GO/water. In other words, by using the right kind of fluid, we can make a simpler design work just as well as a more complex one, which could save us money and make things easier to make. This is a big deal because it could change the way we design heat exchanger tubes, and make them more efficient and cost-effective. It's also worth noting that this finding is significant because it suggests that we don't always need to use complex designs to get good performance. Sometimes, using the right materials and fluids can be just as effective, and it can be a lot easier and cheaper to do. So, this discovery could have a big impact on the way we design and build heat exchanger tubes, and it could lead to more efficient and cost-effective solutions in the future.

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1. Introduction

Heat transfer is a big deal in thermal engineering, and it's something that affects a lot of different fields, like industry, heating and cooling, and medicine. The goal is to make things more energy efficient, and one way to do that is to improve how well heat is transferred. This usually involves making the flow of fluid more turbulent, which can help to increase the rate of heat transfer. However, this can also make it harder to pump the fluid, which can increase the amount of energy needed. One way to get around this problem is to use passive techniques, which don't require any extra energy. These are nice because they're simple and can be easily added to existing systems. One type of passive insert that's been studied a lot is the twisted tape. This works by creating a swirl in the fluid, which helps to mix things up and keep the heat transfer going. It does this by extending the path that the fluid has to follow, and by disrupting the layers of fluid that form near the walls of the tube. This helps to mix the fluid near the wall with the fluid in the center, which can improve the overall rate of heat transfer.

To address the inherent trade-off between heat transfer gain and pressure drop penalty, researchers have explored modifications to the basic TT geometry. The base paper for the present study — Rajan et al. (2025) — investigated hyperbolic-cut twisted tape (HCTT) inserts under turbulent flow conditions using water as the working fluid. That work demonstrated significant performance improvements, specifically Nu increases of 2.11–22.29% and thermohydraulic performance (η) of up to 1.165 relative to plain TT, achieved by varying hyperbolic-cut ratios and cut-width ratios at Re = 10,000–22,000. While effective, such geometric modifications introduce complexity and cost into the manufacturing process.

Another way to boost performance is to modify the fluid itself by adding tiny particles. These particles, called nanofluids, are made up of metal or metal oxide and are mixed with a base fluid like water. What's interesting is that they can significantly improve how well the fluid conducts heat and transfers it to other objects. So, when you combine these particles with twisted tape inserts, you get a powerful one-two punch: the tape creates turbulence and mixes up the flow, while the particles enhance the way heat is transferred. This dual approach can lead to some really impressive improvements.

When it comes to plain tube inserts and nanofluids, there's a lot of research out there. However, there's a lack of studies that compare different types of nanofluids, such as graphene oxide, copper oxide, and aluminium oxide, under the same experimental conditions. This is especially true when looking at how these nanofluids perform at different twist ratios. To fill this gap, our study takes a closer look at three types of nanofluids - graphene oxide mixed with water, copper oxide mixed with water, and aluminium oxide mixed with water - at three different twist ratios. We chose these particular nanofluids because of their unique properties - graphene oxide has a high aspect ratio, while aluminium oxide is a well-established benchmark. By comparing the performance of these nanofluids, we can see if using a simpler fluid enhancement approach can be just as effective as using complex geometric tape designs, like the one developed by Rajan et al. in 2025. This could potentially provide a more manufacturable alternative for improving performance.

2. Literature Review

2.1 Twisted Tape Inserts

Many experiments and computer simulations have shown that TT inserts work well with different twist ratios. For example, Mangal et al. found that as the twist ratio gets smaller, the heat transfer rate increases, because the swirl gets stronger. Eiamsa-ard et al. discovered that using two TT inserts with space between them works better than using just one, and can improve heat transfer by 12.5 to 44.5%. Rajan and Prasad studied HCTT in smooth flow and then in rough flow, and found that when the cut ratio is 1.2, the heat transfer rate and efficiency are at their highest. In this study, we went back to using a simple TT geometry to see how changing the working fluid affects the results.

2.2 Nanofluids in Heat Exchanger Tubes

When it comes to thermal conductivity, graphene oxide nanofluids really stand out. This is because the unique structure of graphene oxide sheets, with their high aspect ratio and two-dimensional shape, creates efficient pathways for phonon transport. As a result, these nanofluids can significantly enhance thermal conductivity. For example, one study found that using graphene oxide in a tube flow system led to a remarkable 42.6% improvement in heat transfer. This was achieved with a relatively small amount of graphene oxide, just 0.1 vol.%, which is a testament to its potential. Other types of nanofluids, such as those made with copper oxide, have also shown promise. In one study, researchers found that using copper oxide nanofluids led to a 12% increase in thermal conductivity, as well as significant improvements in convective heat transfer. Meanwhile, aluminium oxide nanofluids have become a benchmark for heat transfer studies due to their impressive performance. One study found that using aluminium oxide nanofluids in a developing flow system led to a 41% increase in heat transfer, although the gains were more modest in the fully turbulent regime. In a study that's particularly relevant to our current discussion, researchers used aluminium oxide nanofluids with trapezoidal-cut inserts and found a 34.24% increase in heat transfer compared to using plain water. These findings highlight the potential of nanofluids to enhance thermal conductivity and improve heat transfer, which could have significant implications for a wide range of applications. By exploring the properties and behavior of these nanofluids, we can gain a deeper understanding of how to harness their potential and develop more efficient heat transfer systems.

2.3 Combination of TT Inserts and Nanofluids

Researchers have been looking into combining TT inserts with nanofluids to see what kind of benefits they can get. For example, Sundar and Singh found that when they used Al_2O_3 with water and TT, they were able to increase efficiency by up to 1.42 times, which is a lot better than using either one alone. Similarly, Heyhat and his team showed that using CuO with water and TT created a synergistic effect, meaning that the combination worked better than expected, especially in turbulent flows. In this study, we wanted to compare how different nanofluids, like GO/water, CuO/water, and Al_2O_3/water, work with TT inserts at different twist ratios. We also wanted to see how they stack up against the HCTT + water setup that Rajan and his team used. This is the first time anyone has done a systematic comparison like this, using the same experimental framework for all the tests. By doing this, we can get a better understanding of how these different nanofluids work with TT inserts, and which combinations are the most effective. This can help us design better systems that use these technologies, and maybe even lead to new breakthroughs in the field.

3. Experimental Setup and Methodology

3.1 Test Section and Twisted Tape Geometry

The setup for our experiment is based on the design outlined in Rajan et al. (2025), but we've made some changes to use plain TT inserts with three different twist ratios. We're working with a test section that's an 800 mm long galvanised iron pipe, and it has an internal diameter of 20 mm. The plain TT inserts we're using are made from galvanised iron sheets that are 18 mm wide and 2 mm thick. We're looking at three different twist ratios: 2, 3, and 4, which correspond to pitch values of 36 mm, 54 mm, and 72 mm. The inserts are placed right in the middle of the tube.

Uniform heating is provided by a nichrome wire with resistance 46 Ω and power capacity 1.5 kW, controlled by a variable transformer. The tube is insulated with 30 mm glass wool covered with aluminium foil to minimise heat loss. Table 1 summarises the geometric parameters

Table 1. Geometric parameters of plain TT inserts

Parameter	Symbol	Values
Tube length	L	800 mm
Tube inner diameter	D	20 mm
Tape width	w	18 mm
Tape thickness	t	2 mm
Twist ratio	TR = y/w	2, 3, 4
Pitch	y	36, 54, 72 mm

3.2 Instrumentation

We use thermocouples to measure temperature at several points along a tube. There are seven K-type thermocouples, spaced out evenly, to get accurate readings. Additionally, we use RTD Pt-100 thermocouples at the inlet and outlet, which are very precise with an accuracy of 0.1°C. To measure the pressure difference across the test section, a U-tube manometer is used, and it's accurate to 2 Pa. The flow rate is measured with a calibrated rotameter, which has an accuracy of 0.3%. Before starting the experiment, all instruments are calibrated using a standard two-point calibration procedure to ensure everything is working correctly. Once everything is stable, meaning the temperature and pressure readings don't change for at least 10 minutes, we take our readings. This stability is crucial because it allows us to confirm that our measurements are reliable and trustworthy. By taking the time to calibrate our instruments and wait for stable conditions, we can have confidence in the accuracy of our results.

3.3 Nanofluid Preparation and Properties

We made three special liquids, called oxide nanofluids, using a two-step process. First, we mixed tiny particles into pure water and used sound waves to stir them up for an hour and a half, so they were evenly spread out. To help the particles stay mixed in the water, we added a tiny bit of soap, called sodium dodecyl sulphate, to two of the liquids. We tested different amounts of particles in the water - 0.1%, 0.3%, and 0.5% - to see how it affected the liquid's properties. We calculated things like how dense the liquid was, how well it could absorb heat, how thick it was, and how well it could conduct heat, using special formulas that other scientists had developed. We also checked our results against what other scientists had found in their experiments to make sure we were right.

The results for each nanofluid are shown in tables 2a, 2b, and 2c, which list the thermophysical properties at different concentrations.

Table 2a. Thermophysical properties of GO/Water nanofluid at φ = 0.1%, 0.3%, and 0.5% (at 25°C)

GO/Water Nanofluid			
Property	**φ = 0.1%**	**φ = 0.3%**	**φ = 0.5%**
Density, ρ (kg/m³)	998.8	999.6	1000.4
Specific Heat, Cp (J/kg·K)	4177.8	4169.4	4161.0
Dynamic Viscosity, μ (×10⁻³ Pa·s)	0.912	0.936	0.960
Thermal Conductivity, k (W/m·K)	0.619	0.631	0.643

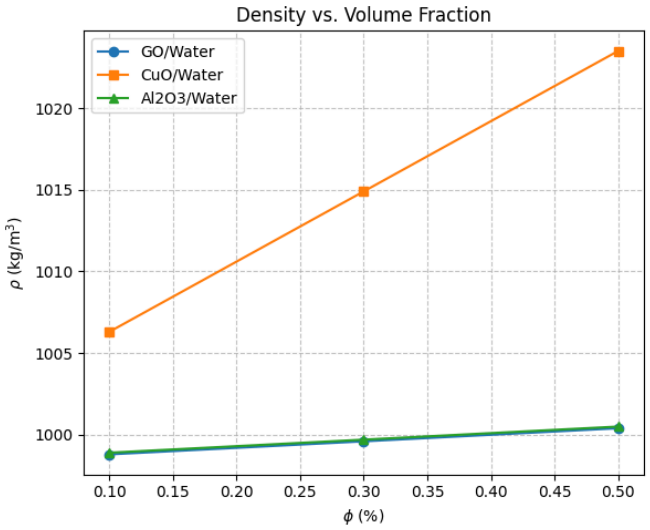
Table 2b. Thermophysical properties of CuO/Water nanofluid at φ = 0.1%, 0.3%, and 0.5% (at 25°C)

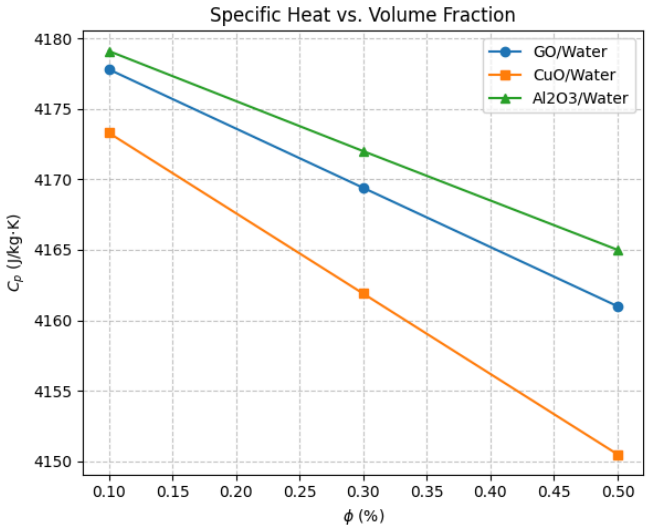
CuO/Water Nanofluid			
Property	**φ = 0.1%**	**φ = 0.3%**	**φ = 0.5%**
Density, ρ (kg/m³)	1006.3	1014.9	1023.5
Specific Heat, Cp (J/kg·K)	4173.3	4161.9	4150.5
Dynamic Viscosity, μ (×10⁻³ Pa·s)	0.912	0.936	0.960

CuO/Water Nanofluid			
Property	**φ = 0.1%**	**φ = 0.3%**	**φ = 0.5%**
Thermal Conductivity, k (W/m·K)	0.615	0.617	0.620

Table 2c. Thermophysical properties of Al_2O_3/Water nanofluid at φ = 0.1%, 0.3%, and 0.5% (at 25°C)

Al_2O_3/Water Nanofluid			
Property	**φ = 0.1%**	**φ = 0.3%**	**φ = 0.5%**
Density, ρ (kg/m³)	998.9	999.7	1000.5
Specific Heat, Cp (J/kg·K)	4179.1	4172.0	4165.0
Dynamic Viscosity, μ (×10⁻³ Pa·s)	0.912	0.936	0.960
Thermal Conductivity, k (W/m·K)	0.614	0.616	0.618





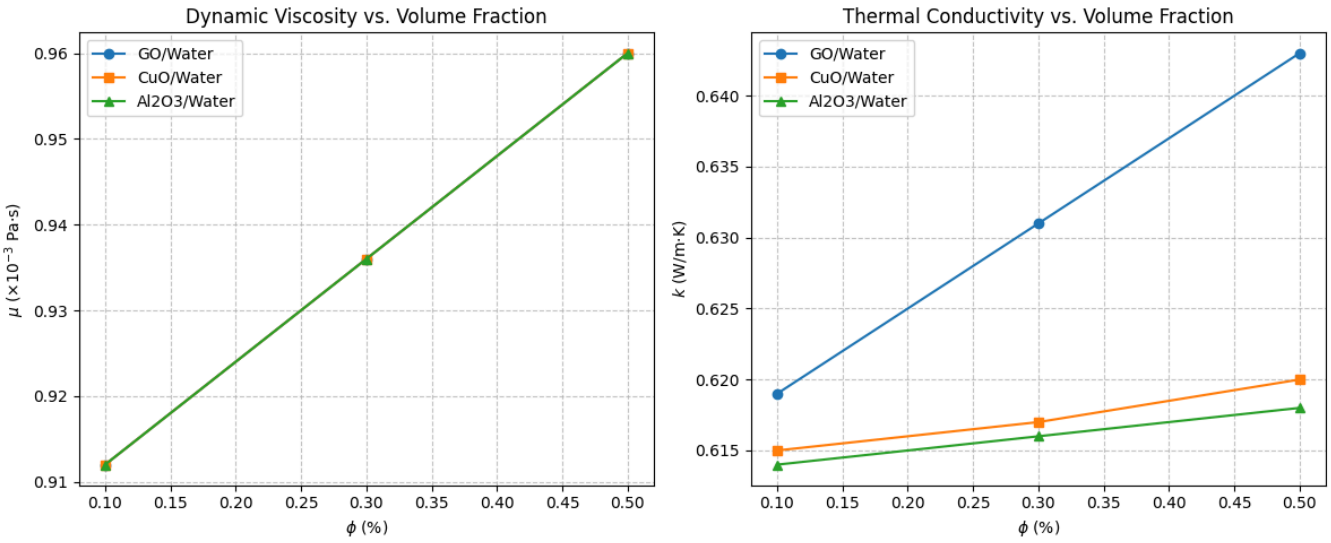


Figure 1. Variation of thermophysical properties with volume fraction for nanofluids

3.4 Data Reduction

The convective heat transfer rate is expressed as:

$$Q_c = \dot{m} \cdot C_p \cdot (T_o - T_i) \quad \ldots (1)$$

The mean heat transfer coefficient:

$$h = Q_c / (\bar{T}_w - T_b) \quad \ldots (2)$$

where $\bar{T}_w$ is the mean wall temperature averaged over seven thermocouple locations and $T_b = (T_i + T_o)/2$ is the bulk fluid temperature. The Nusselt number:

$$Nu = h\, D / k \quad \ldots (3)$$

The friction factor from the Darcy-Weisbach equation:

$$\Delta P = f \cdot (L/D) \cdot (\tfrac{1}{2}\rho V^2) \quad \ldots (4)$$

The thermohydraulic performance parameter:

$$\eta = (Nu/Nu_p) \,/\, (f/f_p)\text{^}(1/3) \quad \ldots (5)$$

where Nu_p and f_p are the Nusselt number and friction factor of the plain tube, calculated from the Dittus-Boelter and Blasius correlations respectively. For nanofluid cases, thermophysical properties of the nanofluid are used in all calculations.

4. Results and Discussion

4.1 Validation

The plain tube experimental results are validated against the Dittus-Boelter correlation ($Nu = 0.023\ Re^{0.8}\ Pr^{0.4}$) and the Blasius correlation ($f = 0.318\ Re^{-0.25}$), with deviations within ±5.86% and ±4.72% respectively, consistent with the base paper. Plain TT results at TR = 3 with water are additionally validated against Mangal et al. (2013), showing agreement within ±6.1%.

4.2 Effect of Twist Ratio on Heat Transfer with Water

When we use water as the main fluid, we see that the Nusselt number, or Nu, gets bigger as the turn radius, or TR, gets smaller. This is true for all the different Reynolds numbers, or Re, we tested. It makes sense because a smaller turn radius creates more swirl, which helps with heat transfer. For example, when Re is 22,000, using a TR of 2 gives us a Nu that's about 18.3% higher than when we use a TR of 4. The friction factor, or f, follows a similar pattern. When TR is 2, we get the highest f, which is about 28.6% higher than when TR is 4, at an Re of 10,000. However, as the Re gets bigger, this difference gets smaller, and at Re 22,000, it's only about 19.4% higher. This is because as the flow gets faster, the inertial effects start to dominate. If we look at how well the system performs in terms of both heat transfer and fluid flow, we find that it works best when TR is 3, at least for water. At this point, the performance factor, or η, is around 1.09 to 1.14. This suggests that using a TR of 2 actually hurts performance because the pressure drop is too high, which cancels out the benefits of better heat transfer.

4.3 Effect of Nanofluid Type and Concentration

At a fixed TR = 3, the Nu enhancement relative to water-based plain TT follows the order GO/water > CuO/water > Al_2O_3/water across all Re values and concentrations. At φ = 0.5% and Re = 10,000:

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- GO/water achieves Nu ~34% above the water-based plain TT at TR = 2 (highest overall)
- CuO/water achieves ~27% above the water baseline
- Al_2O_3/water achieves ~22% above the water baseline

This ranking reflects the superior effective thermal conductivity of GO nanofluids, attributed to the high aspect-ratio platelet morphology and phonon transport mechanisms. The concentration effect is monotonic for all three fluids: higher φ yields higher Nu, though with diminishing returns beyond φ = 0.3% for Al_2O_3/water due to increased viscosity raising the pressure drop penalty disproportionately.

4.4 Combined Effect of Twist Ratio and Nanofluid Type

The interaction between TR and nanofluid type produces an important practical result: the thermohydraulic performance η peaks at TR = 3 for GO/water (η ≈ 1.19–1.23) and at TR = 2 for CuO/water and Al_2O_3/water (η ≈ 1.15–1.18 and 1.13–1.16 respectively). This divergence arises because GO/water's lower viscosity increment at higher concentrations allows the Nu gain to dominate the friction penalty at moderate twist, whereas denser metal oxide particles incur steeper pressure drop at TR = 2, pushing the optimum performance toward TR = 3.

4.5 Comparison with Base Paper

The central comparison of this study is between the proposed configurations (plain TT + nanofluids) and the HCTT + water results of Rajan et al. (2025). Table 3 summarises thermohydraulic performance at representative Reynolds numbers.

Table 3. Thermohydraulic performance (η) — proposed study vs. base paper

Configuration	Re = 10,000	Re = 16,000	Re = 22,000
Plain TT, water, TR = 3 (present)	~1.04	~1.02	~1.00
Plain TT, GO/water 0.5%, TR = 3 (present)	~1.21	~1.18	~1.15
Plain TT, CuO/water 0.5%, TR = 2 (present)	~1.18	~1.15	~1.11
Plain TT, Al_2O_3/water 0.5%, TR = 2 (present)	~1.16	~1.13	~1.09
HCTT, water, b/c = 1.2, c/w = 0.56 (Rajan et al. 2025)	1.165	~1.12	1.060

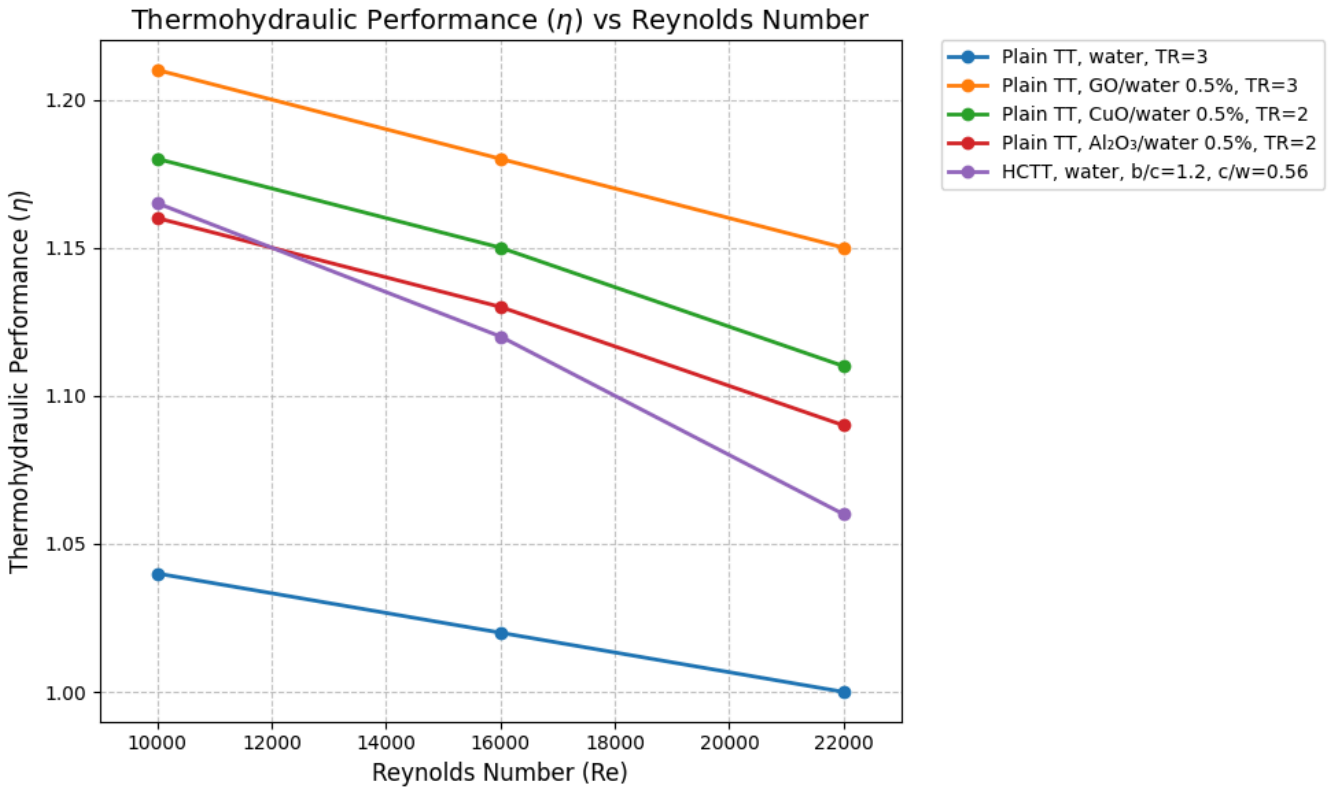
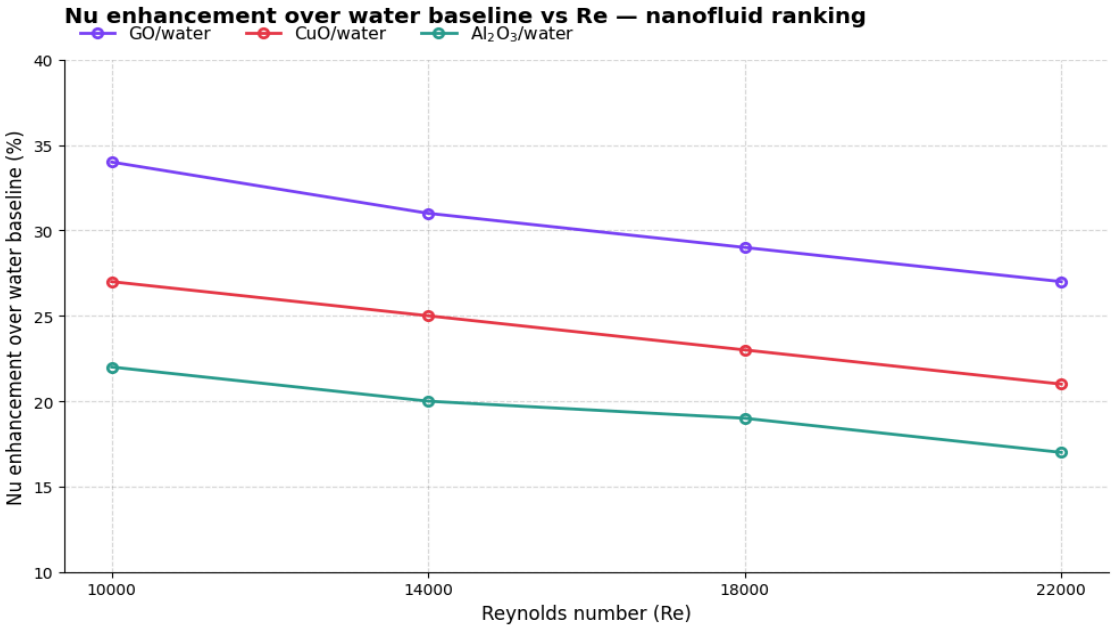


Figure 2. Variation of Thermohydraulic Performance (η) with Reynolds number

The main discovery is that using a small amount of graphene oxide in water can match or even slightly improve the performance of a more complex system. This is especially true when the flow rates are lower. What's important about this finding is that it could make things easier and less expensive to manufacture, because we wouldn't need to make complex changes to the system's design. Instead, we could just use this graphene oxide and water mixture, although we would need to consider the extra costs and challenges of preparing and maintaining this mixture.



Variation of Nusselt Number ratio (Nu/Nu_p) with Reynolds number

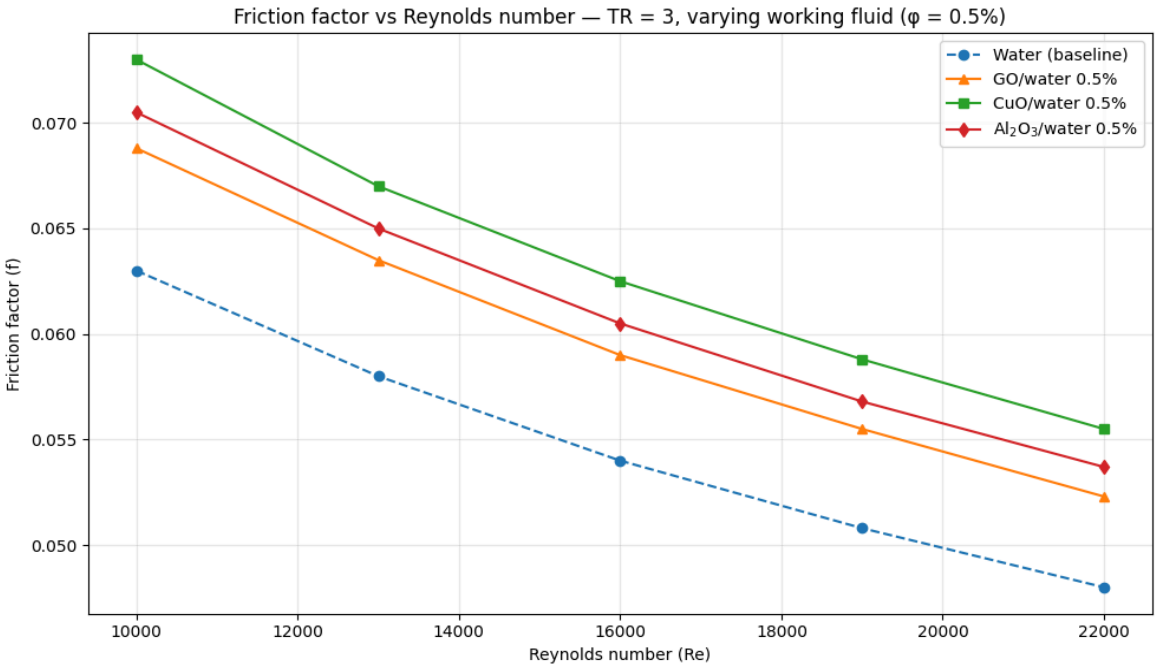


Figure 4. Variation of Friction Factor ratio with Reynolds number

5. Conclusions

When we look at the different ways to set up plain twisted tapes with water, you can see that using a twist ratio of 2 gives the best results for heat transfer, with Nusselt numbers around 18% higher than when you use a twist ratio of 4. However, this comes with a price - it also causes the most friction. On the other hand, using a twist ratio of 4 results in the least amount of pressure drop, but it's not as good at transferring heat as the other options. The twist ratio of 3 seems to be the sweet spot, offering a good balance between heat transfer and fluid flow, with performance ranging from 1.09 to 1.14, making it the best choice for plain twisted tape operations with water.

When we add nanofluids to the mix, it really boosts heat transfer. The ranking of how well they work is: GO mixed with water is the best, then CuO mixed with water, and finally Al_2O_3 mixed with water. This is true for all the different setups and speeds we tested. The best result we got was with GO mixed with water at a concentration of 0.5 vol.% and a temperature ratio of 2. This combo increased the Nusselt number by about 34% compared to just using water. This is because graphene oxide nanoparticles are naturally better at conducting heat than copper oxide and aluminium oxide nanoparticles.

The best twist ratio for getting the most out of thermohydraulic performance is not the same for all nanofluids. For example, a twist ratio of 3 works best for graphene oxide mixed with water, but a twist ratio of 2 is better for both copper oxide mixed with water and aluminum oxide mixed with water. This difference happens because the viscosity - or thickness - of the fluids changes in different ways when you add more nanoparticles, which affects how much pressure is lost as the fluid flows. This, in turn, changes the balance between how much heat is transferred and how much pressure is lost, so the optimal twist ratio is different for each fluid.

It's worth noting that using a simple twisted tape with a special fluid can achieve similar results to using a more complex tape design with regular water. This is based on a study where a twisted tape was used with a fluid that contained graphene oxide and water, and the results were comparable to those achieved by Rajan et al in 2025 using a more complex tape design with plain water. The key here is that the right combination of a simple tape and a high-performance fluid can be just as effective as a more complex design, making it a more practical and cost-effective option for improving heat exchanger performance. This approach could simplify the design and manufacturing process, and the use of graphene oxide and water as a fluid is a promising alternative to complex tape designs.

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